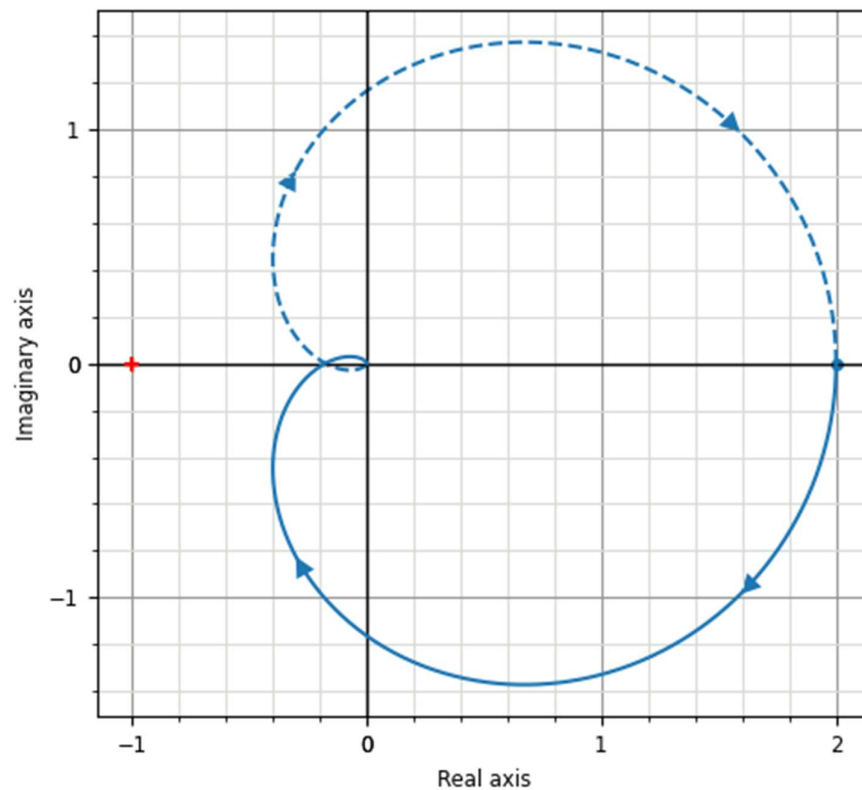
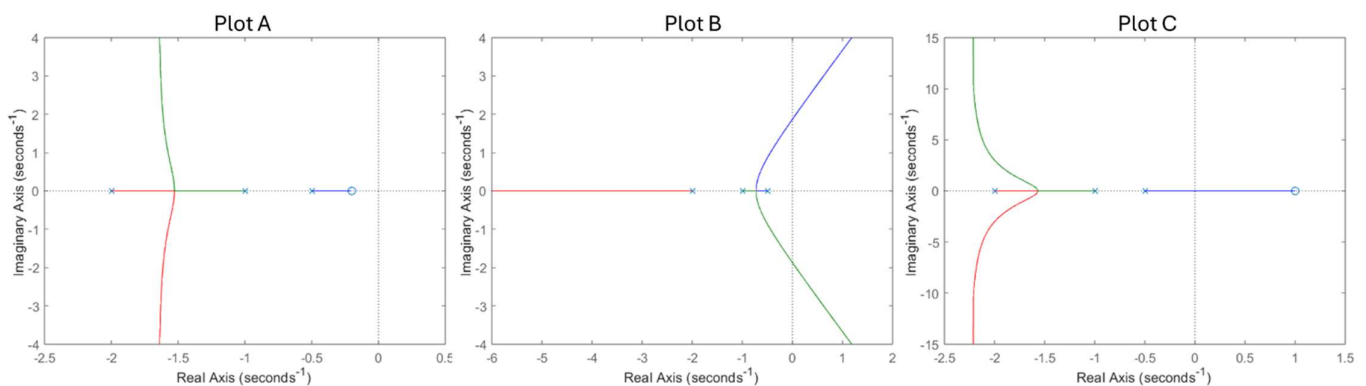


Tut Test 4 [10 marks]

The Nyquist plot of an unknown third-order system is shown below. The system is stable in open loop.



A) Three possible root locus plots for the system are shown below. Which one do you believe matches the system? Explain your reasoning with reference to the Nyquist plot. [3 marks]



B) Using the Nyquist plot, confirm that the DC gain of the uncontrolled open-loop system is 2. (You may draw on the plot to support your answer) [2 marks]

C) If you placed this system in negative feedback with a proportional controller, would it be possible to achieve tracking of a constant setpoint with 90% accuracy? Explain. [5 marks]

Tut Test 4

Answer sheet

Name: _____

Student No: _____

a)

b)

c)